

Matthew Smith

Senior Software Engineer | Federal & Private Sector Experience

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SUMMARY

Senior Software Engineer specializing in building, modernizing and delivering secure, scalable enterprise systems across federal and private sectors. Experienced in Java (Spring Boot), C# (.NET), and Python (Django) and modern TypeScript frameworks, with deep involvement in mission-critical FEMA platforms including PrepToolkit and URT. Proven track record designing RESTful APIs, leading legacy modernization initiatives, delivering containerized services in production environments, and mentoring engineering teams. Strong focus on maintainability, performance, and clean architecture.

CORE EXPERTISE

Backend: Java (Spring Boot), C# (.NET), Python (Django), REST APIs, MySQL, PostgreSQL, SQL Server, Flyway

Frontend: TypeScript, Svelte, React, Angular, GWT, Bootstrap, Tailwind

System Design: Microservices, API-Driven Design, Event-Driven Patterns, Distributed Session Management, Layered Systems

DevOps & Tools: Docker/Podman, Gradle, Maven, AWS (Lambda, S3), Git, Apache Tomcat

Platforms: Liferay 7, Node.js, npm, Cloudflare Workers & Pages

PROFESSIONAL EXPERIENCE

Senior Software Engineer | 4 Arrows Consulting, Inc. • Somerset, KY (Remote) • 2019 – Present

- **Serverless Platform Architecture:** Designed and delivered the Svelte frontend for a TypeScript monorepo of AWS serverless applications (Lambda, API Gateway, CloudFront, ElastiCache, RDS) serving as a unified access portal for FEMA platforms including PrepToolkit, RTLT, RIS, and OneResponder.
- **System Architecture & Presence:** Spearheaded the implementation of a real-time presence tracking system for the Unified Reporting Tool (URT). Independently designed and optimized distributed session logic (polling/timeouts) to ensure data integrity for concurrent federal users.
- **Cross-Stack Feature Ownership:** Delivered complex features across backend (Spring Boot/.NET), database, and frontend layers within FEMA preparedness platforms (PrepToolkit, URT), ensuring cohesive integration between APIs, data models, and user interfaces.
- **Operational Excellence:** Automated enterprise reporting workflows (Docmosis/Apache POI), reducing processing time from hours to minutes and directly improving operational efficiency for national exercise programs.
- **Stakeholder Alignment:** Translates complex FEMA business requirements (THIRA/SPR, CPG, NIMS, UAWG) into scalable technical solutions, coordinating closely with project leads to ensure all deliverables meet federal compliance and security standards.
- **Development Workflow:** Consistently utilizes containerized environments (Docker/Podman) and Flyway-based database migrations across projects, including scripted Podman-based Liferay container management to streamline local development and module redeployment. Conducts peer code reviews to reinforce clean architecture and maintainable code practices.
- **Developer Experience:** Standardized editor configurations across VS Code, IntelliJ, and Visual Studio to enforce consistent code formatting team-wide, eliminating noise commits caused by style-only changes.

Tech Stack: Java, Spring Boot, Gradle, GWT, MySQL, Liferay 7, Apache HTTP Server, JavaScript/TypeScript, Svelte, React, Node.js, npm, AWS Lambda, Git, Podman, Docmosis, OpenCSV, Apache POI

Applications Developer | United Parcel Service (UPS) • Louisville, KY • 2017 – 2019

- **Mission-Critical Reliability:** Engineered backend logic and UI features for real-time transportation visibility systems, ensuring operational uptime in a 24/7 high-throughput production environment.
- **Business Logic Abstraction:** Architected complex decision-making flows by translating high-level business requirements into discrete rules for the Drools inference engine (BRMS), reducing manual intervention and decoupling policy from code.
- **Performance Engineering:** Conducted deep-dive bottleneck analysis and performance tuning (JMeter), identifying and resolving critical latency issues under peak traffic conditions.

Tech Stack: Java, Spring Framework, Maven/Gradle, Drools (BRMS), JMeter, Roadrunner, WebLogic, ActiveMQ, SQL Server, TFVC

Software Developer | Apax Software • Lexington, KY • 2016 – 2017

- **Modular Architecture:** Designed reusable backend components and modular UI features, significantly accelerating release cycles and ensuring scalable feature expansion across multiple client projects.
- **Technical Consultation:** Partnered directly with stakeholders (Presbyterian Church (USA), University of Kentucky) to translate vague business needs into production-ready technical specifications and RESTful architectures.
- **Security & Governance:** Implemented robust server-side validation, authentication, and role-based access control (RBAC) for sensitive event registration and CMS workflows.

Tech Stack: Python, Django, JavaScript (AngularJS, jQuery), PostgreSQL, Bootstrap, AWS, Git, PHP, Drupal

Programming Analyst | Knowledge Facilitations Group (KFG) • Richmond, KY • 2015 – 2016

- **Data Pipeline Engineering:** Developed Python-based transformation engines to validate and map disparate industry standards (X12, EDIFACT, XML, JSON), ensuring seamless data exchange across the EDITRACE platform.
- **Operational Observability:** Built real-time monitoring dashboards that provided immediate insight into automated data pipelines, reducing troubleshooting time for critical EDI processes.

Tech Stack: Python, EDI (X12, XML, EDIFACT, Tradacoms, JSON, IDoc), Bootstrap, Git

PROJECTS

RAG Document Q&A Demo • <https://github.com/msmith1392/rag-demo>

- Locally-running offline RAG system for document Q&A using sentence-transformers for embeddings and FAISS for vector search.

Tech Stack: Python, LangChain, FAISS, Mistral 7b

Personal Portfolio Site • <https://github.com/msmith1392/msmith1392.github.io>

- Production-deployed single-page portfolio hosted on GitHub Pages.

Tech Stack: React, TypeScript, Tailwind, Vite

EDUCATION

B.S. in Computer Science, 2015 | Eastern Kentucky University, Richmond, KY

Member, Upsilon Pi Epsilon Honors Society for the Computing and Information Disciplines